INDIAN APSRTC BUS STATIONS & TRAVELLING

Title : INDIAN APSRTC BUS STATIONS & TRAVELLING

Submitted By : M SravanKumar

Date : 5 -5-2026

Batch : (BN007)

Data Analysis Tools used : Power BI,Excel,Mysql,python

INDEX

Indian APSRTC Bus Data Analytics Project

Abstract

This project presents a comprehensive data analytics study on the operations of the Indian Andhra Pradesh Road Transport Corporation (APSRTC) bus system. The objective is to analyze large-scale transportation data to uncover insights related to passenger demand, revenue generation, operational efficiency, and profitability. A structured dataset was designed using a star schema consisting of one fact table and multiple dimension tables, enabling efficient data modeling and analysis.

Data preprocessing and cleaning were performed using MySQL and Python, where missing values, duplicates, and inconsistencies were handled to ensure high data quality. Exploratory Data Analysis (EDA) was conducted using Python libraries to identify trends, patterns, and relationships among key variables such as passengers, revenue, delays, and fuel costs. Various visualization techniques were applied, including line charts, bar charts, scatter plots, and advanced visualizations such as donut, funnel, and ribbon-style charts.

An interactive business intelligence dashboard was developed using Microsoft Power BI to present insights in a user-friendly manner. The dashboard includes multiple analytical views such as executive overview, operational performance, financial analysis, and customer behavior. Key performance indicators (KPIs) such as total revenue, total profit, occupancy rate, delay metrics, and profit margin were calculated using DAX measures.

The analysis highlights that passenger demand varies significantly across routes, cities, and time periods, with notable increases during festival seasons. Operational challenges such as delays are strongly influenced by traffic conditions and impact overall service efficiency. Additionally, premium bus types contribute more to profitability, while fuel costs play a critical role in determining margins.

This project demonstrates an end-to-end data analytics workflow, integrating data engineering, statistical analysis, and visualization to support data-driven decision-making in public transportation systems. The findings can assist stakeholders in optimizing operations, improving service quality, and enhancing customer satisfaction.

Introduction

Public transportation plays a vital role in the economic and social development of a country, especially in a diverse and populous nation like India. Road Transport Corporations (RTC) are responsible for providing affordable and accessible bus services across urban and rural regions. With the increasing demand for efficient and reliable transportation, it has become essential to leverage data analytics to improve operational performance, optimize resources, and enhance passenger satisfaction.

The rapid growth of data generated from transportation systems—such as trip details, passenger counts, revenue, delays, and operational factors—provides an opportunity to extract meaningful insights. However, raw data in its original form is often unstructured and inconsistent, making it difficult to analyze directly. This project addresses these challenges by applying data analytics techniques to transform raw RTC bus data into actionable insights.

The primary aim of this project is to analyze APSRTC bus operations using a structured data model and modern analytical tools. A star schema data model was designed, consisting of a central fact table containing transactional trip data and multiple dimension tables representing stations, buses, routes, and dates. This structure enables efficient querying and supports multidimensional analysis.

The project follows a systematic approach, beginning with data cleaning and preprocessing using MySQL and Python to ensure data quality. Exploratory Data Analysis (EDA) is then performed to identify trends, patterns, and anomalies in key variables such as passenger demand, revenue, delays, and fuel costs. Advanced visualization techniques are used to present findings in an intuitive and interactive manner.

Furthermore, an interactive dashboard is developed using Microsoft Power BI to provide a comprehensive view of system performance. The dashboard integrates key performance indicators (KPIs) and visual analytics to support data-driven decision-making. It enables stakeholders to monitor operational efficiency, evaluate financial performance, and understand customer behavior.

Overall, this project demonstrates the application of data analytics in the transportation domain, highlighting how data-driven insights can be used to improve service quality, optimize operations, and support strategic planning in APSRTC bus systems.

Problem Statement

The Indian Andhra Pradesh Road Transport Corporation (APSRTC) operates a large-scale bus transportation system that generates vast amounts of data related to trips, passengers, revenue, routes, and operational conditions. However, this data is often underutilized due to a lack of structured analysis and effective visualization tools. As a result, decision-making related to route optimization, resource allocation, and service improvement remains inefficient and largely experience-based rather than data-driven.

One of the key challenges faced by APSRTC systems is the inability to accurately analyze passenger demand patterns across different routes, cities, and time periods. Additionally, operational issues such as delays, cancellations, and fluctuating occupancy rates impact service quality and customer satisfaction. Financial performance is also affected by factors such as fuel costs, inefficient route planning, and underperforming services.

Another significant problem is the absence of an integrated system that combines data from multiple sources into a unified structure for analysis. Without proper data cleaning, transformation, and modeling, inconsistencies such as missing values, duplicate records, and inaccurate entries further reduce the reliability of insights.

Therefore, there is a need to develop a comprehensive data analytics solution that can:

- Clean and organize large volumes of transportation data
- Analyze key performance indicators such as revenue, profit, passenger volume, and delays
- Identify patterns and trends in passenger demand and operational efficiency
- Provide interactive and visual dashboards for better decision-making

This project aims to address these challenges by implementing a structured data model, performing in-depth analysis, and developing an interactive dashboard to support data-driven decisions in the RTC bus transportation system.

Objectives of the Project

Indian APSRTC Bus Data Analytics

The primary objective of this project is to apply data analytics techniques to APSRTC bus transportation data in order to derive meaningful insights that support operational efficiency, financial performance, and customer satisfaction.

The specific objectives of the project are as follows:

- To design a structured data model using a star schema consisting of fact and dimension tables for efficient data analysis
- To clean and preprocess raw data by handling missing values, duplicates, and inconsistencies using MySQL and Python
- To analyze passenger demand across different routes, cities, and time periods
- To evaluate financial performance by examining key metrics such as revenue, profit, and profit margin
- To assess operational efficiency by analyzing delays, cancellations, and occupancy rates
- To identify patterns and trends influenced by factors such as traffic conditions, weather, and festival periods
- To create calculated metrics (KPIs) such as revenue per passenger, profit margin, and average occupancy
- To perform exploratory data analysis (EDA) using Python for deeper insights and visualization
- To develop an interactive and user-friendly dashboard using Microsoft Power BI for effective data visualization
- To enable data-driven decision-making for improving route planning, resource allocation, and service quality

Overall, the project aims to demonstrate how data analytics can be leveraged to transform raw transportation data into actionable insights for better management of RTC bus operations.

Scope of the Project

The scope of this project focuses on analyzing APSRTC bus transportation data to derive meaningful insights related to operational performance, financial outcomes, and passenger demand patterns. The analysis is carried out using a structured dataset and modern data analytics tools to support informed decision-making.

This project covers the design and implementation of a star schema data model, consisting of one fact table and multiple dimension tables, to enable efficient data organization and querying. The scope includes data cleaning and preprocessing using MySQL and Python, where issues such as missing values, duplicate records, and inconsistencies are addressed to ensure data accuracy and reliability.

The analytical scope involves evaluating key performance indicators (KPIs) such as total revenue, total profit, passenger volume, occupancy rate, delays, and cancellations. It also includes time-based analysis (daily, monthly, yearly trends), route-wise and city-wise performance analysis, and the impact of external factors such as traffic conditions, weather, and festival periods on operations.

Visualization is a key component of the project, where insights are presented through charts and interactive dashboards. A multi-page dashboard is developed using Microsoft Power BI to provide a comprehensive view of executive, operational, financial, and customer-related metrics.

However, the scope of this project is limited to historical data analysis and descriptive analytics. It does not include real-time data integration, advanced predictive modeling, or optimization algorithms. The dataset used is synthetic and intended for analytical and learning purposes, which may not fully represent real-world complexities.

Overall, the project demonstrates an end-to-end data analytics workflow, from data preparation to visualization, within the defined scope of descriptive and exploratory analysis in the RTC bus transportation domain.

Dataset Overview

The dataset used in this project represents the operational and financial data of the Indian Andhra Pradesh Road Transport Corporation (APSRTC) bus system. It is designed using a **star schema structure**, which consists of one central fact table and multiple dimension tables. This structure enables efficient data organization, faster querying, and multidimensional analysis.

The dataset contains approximately 500KB of data and includes detailed information related to trips, passengers, revenue, routes, buses, stations, and time. It is suitable for performing descriptive and exploratory data analysis.

Fact Table: fact_trips

The fact table contains transactional data at the trip level and serves as the core of the dataset. Each record represents a single bus trip with measurable metrics.

Key Columns:

- trip_id – Unique identifier for each trip
- date_id – Reference to date dimension
- bus_id – Reference to bus dimension
- route_id – Reference to route dimension
- station_id – Reference to station dimension
- passengers – Number of passengers per trip
- revenue – Total revenue generated
- fuel_cost – Cost of fuel per trip
- delay_min – Delay time in minutes
- cancellations – Number of cancellations
- occupancy_rate – Seat occupancy percentage
- ticket_price_avg – Average ticket price
- profit – Profit per trip
- weather – Weather condition during trip
- traffic_level – Traffic condition (Low/Medium/High)
- festival_flag – Indicates festival period (0/1)

trip_id	date_id	bus_id	route_id	station_id	passengers	revenue	fuel_cost	delay_min	cancellations	occupancy_rate	ticket_price_avg	profit	weather	traffic_level	festival_flag
1	1	117	127	35	32	15061.01	4614.18	114	0	0.97	410.41	6301.82	Rain	Medium	0
2	2	130	140	67	25	16776.19	632.15	39	0	0.51	402.88	8149.27	Fog	Low	0
3	3	88	73	34	34	16987.56	4107.37	77	0	0.7	368.67	8263.1	Rain	High	1
4	4	72	2	33	48	11761.59	2257.32	60	0	0.96	224.94	8565.96	Fog	Medium	1
5	5	139	8	46	11	15821.4	3852.16	94	1	0.63	682.54	2428.21	Clear	Medium	1
6	6	64	139	81	47	10355.51	4768.37	5	1	0.54	571.16	1080.64	Rain	High	1
7	7	72	124	45	50	19079.07	1878.29	24	0	0.46	524.19	8443.32	Rain	Medium	1
8	8	21	53	19	48	4623.27	903	102	1	0.76	412.8	901.83	Fog	Low	0
9	9	18	10	94	43	18396.72	1545.34	45	0	0.82	371.97	2528.16	Clear	Medium	1
10	10	64	112	87	49	5912.08	1499.42	30	1	0.52	650.52	5498.89	Clear	Medium	0
11	11	82	8	15	52	2951.74	3838.32	101	0	0.8	468.78	8626.62	Clear	Medium	1
12	12	110	36	78	34	9342.98	778.36	79	1	0.94	525.36	9019.42	Clear	Low	0
13	13	95	106	66	45	12683.43	729.2	62	1	0.55	170	6466.48	Fog	Low	0
14	14	166	14	39	25	5388.81	3621.24	90	1	0.47	430.04	9684.21	Rain	Medium	0
15	15	32	86	18	27	11560.13	3005.24	93	0	0.9	191.22	7434.46	Fog	Low	0
16	16	11	149	70	22	16349.97	2379.95	67	1	0.95	137.68	5513.38	Clear	Medium	0
17	17	45	50	59	46	5884.61	3089.16	38	1	0.53	582.51	1776.52	Rain	High	1
18	18	33	20	32	36	12026.12	3808.62	79	0	0.83	292.37	4573.46	Rain	High	1
19	19	102	87	21	43	7446.94	965.62	35	0	0.75	775.54	8312.24	Clear	Medium	1
20	20	169	135	40	20	11065.6	2989.79	6	0	0.54	523.85	3194.65	Fog	High	1
21	21	136	137	92	10	13671.46	4209.88	84	0	0.89	551.61	5993.58	Fog	Low	0
22	22	11	129	27	22	7658.03	2130.93	65	0	0.68	237.78	9842.63	Clear	Medium	0
23	23	179	38	2	39	19895.17	3933.58	54	1	0.82	190.22	2157.03	Rain	Medium	1
24	24	88	44	97	14	11711.73	4304.39	71	0	0.58	467.79	1463.84	Rain	Medium	1
25	25	41	86	80	27	16664.77	4194.18	41	0	0.89	747.89	8058.17	Fog	Medium	1
26	26	145	44	39	23	14777.57	3426.6	26	1	0.56	139.02	736.76	Rain	Medium	1
27	27	76	74	86	22	17203.3	3728.05	1	0	0.62	455.72	6610.93	Clear	High	0
28	28	121	119	56	41	4302.22	4044.54	69	0	0.63	239.1	3689.58	Rain	Low	1
29	29	46	122	13	16	5228.52	3801.69	3	1	0.52	647.39	7376.23	Fog	High	0
30	30	160	134	56	17	10260.81	2532.56	20	1	0.92	577.36	1642.41	Fog	Low	1
31	31	80	145	66	59	17194.24	1216.04	0	0	0.47	168.04	3635.42	Clear	Low	1
32	32	54	51	78	16	4754.53	1893.79	37	0	0.74	531.44	661.85	Fog	Low	0
33	33	86	38	56	58	19450.25	1432.61	59	1	0.89	788.52	2733.36	Clear	Low	0
34	34	92	51	28	25	3424.44	2126.5	75	0	0.75	469.7	5090.96	Rain	Low	1

Dimension Tables

1. dim_stations

Provides information about bus stations.

- station_id – Unique station identifier
- station_name – Name of the station
- city – City where station is located
- state – State name
- zone – Region classification (North/South/East/West)
- capacity – Station capacity

station_id	station_name	city	state	zone	capacity
1	Station_1	Guntur	Andhra Pradesh	West	31
2	Station_2	Tirupati	Andhra Pradesh	East	13
3	Station_3	Visakhapatnam	Andhra Pradesh	North	41
4	Station_4	Tirupati	Andhra Pradesh	West	37
5	Station_5	Tirupati	Andhra Pradesh	West	46
6	Station_6	Vijayawada	Andhra Pradesh	South	48
7	Station_7	Visakhapatnam	Andhra Pradesh	North	28
8	Station_8	Visakhapatnam	Andhra Pradesh	West	19
9	Station_9	Visakhapatnam	Andhra Pradesh	East	36
10	Station_10	Tirupati	Andhra Pradesh	East	36
11	Station_11	Guntur	Andhra Pradesh	South	28
12	Station_12	Visakhapatnam	Andhra Pradesh	West	45
13	Station_13	Tirupati	Andhra Pradesh	North	16
14	Station_14	Vijayawada	Andhra Pradesh	East	43
15	Station_15	Guntur	Andhra Pradesh	West	6
16	Station_16	Vijayawada	Andhra Pradesh	West	7
17	Station_17	Guntur	Andhra Pradesh	South	41
18	Station_18	Tirupati	Andhra Pradesh	East	21
19	Station_19	Hyderabad	Andhra Pradesh	East	6
20	Station_20	Guntur	Andhra Pradesh	North	6
21	Station_21	Vijayawada	Andhra Pradesh	East	32
22	Station_22	Tirupati	Andhra Pradesh	North	27
23	Station_23	Guntur	Andhra Pradesh	East	41
24	Station_24	Hyderabad	Andhra Pradesh	South	36
25	Station_25	Hyderabad	Andhra Pradesh	East	37

2. dim_buses

Contains details about buses.

- bus_id – Unique bus identifier
- bus_type – Type (AC, Non-AC, Sleeper, etc.)
- operator – Operating authority (APSRTC/TSRTC)
- seats – Total seating capacity
- fuel_type – Diesel/Electric
- rating – Bus rating

bus_id	bus_type	operator	seats	fuel_type	rating
1	Semi-Sleeper	TSRTC	48	Electric	4.26
2	Non-AC	TSRTC	46	Electric	3.16
3	AC	TSRTC	33	Electric	4.75
4	Sleeper	TSRTC	41	Diesel	4.84
5	AC	APSRTC	58	Diesel	3.12
6	AC	APSRTC	38	Diesel	3.55
7	Non-AC	TSRTC	48	Diesel	4.61
8	Semi-Sleeper	APSRTC	41	Electric	4.5
9	Non-AC	APSRTC	38	Diesel	3.37
10	Non-AC	TSRTC	36	Electric	3.42
11	Non-AC	TSRTC	57	Diesel	3.74
12	Non-AC	APSRTC	43	Electric	3.97
13	Sleeper	TSRTC	49	Electric	4.24
14	Semi-Sleeper	APSRTC	48	Diesel	3.74
15	AC	TSRTC	55	Diesel	3.93
16	AC	APSRTC	44	Diesel	4.49
17	Semi-Sleeper	TSRTC	45	Diesel	3.07
18	AC	APSRTC	50	Electric	3.5
19	Semi-Sleeper	APSRTC	34	Diesel	4.43
20	AC	APSRTC	32	Diesel	4.79
21	Non-AC	TSRTC	41	Electric	4.02
22	AC	APSRTC	54	Diesel	4.06
23	Semi-Sleeper	APSRTC	49	Diesel	3.21
24	Semi-Sleeper	TSRTC	56	Diesel	3.89
25	Semi-Sleeper	TSRTC	50	Diesel	4.07
26	Semi-Sleeper	APSRTC	33	Diesel	3.48
27	Semi-Sleeper	APSRTC	50	Electric	3.54
28	Sleeper	TSRTC	52	Diesel	3.75

3. dim_routes

Defines route-related information.

- route_id – Unique route identifier
- source – Starting location
- destination – Ending location
- distance_km – Distance of route
- duration_hr – Travel duration

route_id	source	destination	distance_km	duration_hr
1	Station_54	Station_29	710	12.42
2	Station_87	Station_60	130	4.14
3	Station_57	Station_82	776	4.13
4	Station_1	Station_2	534	8.52
5	Station_63	Station_1	423	9.3
6	Station_54	Station_47	759	9.12
7	Station_55	Station_69	249	2.28
8	Station_40	Station_20	586	13.28
9	Station_15	Station_11	515	4.72
10	Station_21	Station_2	138	2.81
11	Station_47	Station_67	317	13.44
12	Station_73	Station_87	704	14.38
13	Station_53	Station_12	620	13.07
14	Station_9	Station_20	538	12.33
15	Station_74	Station_5	587	10.17
16	Station_52	Station_37	96	8.71
17	Station_57	Station_38	337	2.22
18	Station_26	Station_94	65	6.72
19	Station_41	Station_9	760	6.22
20	Station_35	Station_98	450	4.64
21	Station_63	Station_53	135	11.13
22	Station_25	Station_44	299	7.94
23	Station_90	Station_99	440	2.13
24	Station_75	Station_86	384	4.08
25	Station_38	Station_24	678	10.57
26	Station_2	Station_81	568	2.07
27	Station_7	Station_74	186	12.92
28	Station_82	Station_30	737	7.93

4. dim_dates

Provides time-related attributes for analysis.

- date_id – Unique date identifier
- date – Actual date
- day – Day of month

- month – Month
- year – Year
- weekday – Day name

date_id	date	day	month	year	weekday
1	2023-01-01 00:00:00	1	1	2023	Sunday
2	2023-01-02 00:00:00	2	1	2023	Monday
3	2023-01-03 00:00:00	3	1	2023	Tuesday
4	2023-01-04 00:00:00	4	1	2023	Wednesday
5	2023-01-05 00:00:00	5	1	2023	Thursday
6	2023-01-06 00:00:00	6	1	2023	Friday
7	2023-01-07 00:00:00	7	1	2023	Saturday
8	2023-01-08 00:00:00	8	1	2023	Sunday
9	2023-01-09 00:00:00	9	1	2023	Monday
10	2023-01-10 00:00:00	10	1	2023	Tuesday
11	2023-01-11 00:00:00	11	1	2023	Wednesday
12	2023-01-12 00:00:00	12	1	2023	Thursday
13	2023-01-13 00:00:00	13	1	2023	Friday
14	2023-01-14 00:00:00	14	1	2023	Saturday
15	2023-01-15 00:00:00	15	1	2023	Sunday
16	2023-01-16 00:00:00	16	1	2023	Monday
17	2023-01-17 00:00:00	17	1	2023	Tuesday
18	2023-01-18 00:00:00	18	1	2023	Wednesday
19	2023-01-19 00:00:00	19	1	2023	Thursday
20	2023-01-20 00:00:00	20	1	2023	Friday
21	2023-01-21 00:00:00	21	1	2023	Saturday
22	2023-01-22 00:00:00	22	1	2023	Sunday
23	2023-01-23 00:00:00	23	1	2023	Monday
24	2023-01-24 00:00:00	24	1	2023	Tuesday
25	2023-01-25 00:00:00	25	1	2023	Wednesday
26	2023-01-26 00:00:00	26	1	2023	Thursday
27	2023-01-27 00:00:00	27	1	2023	Friday
28	2023-01-28 00:00:00	28	1	2023	Saturday

Key Characteristics

- Structured in **star schema** for efficient analysis
- Supports **time-based, route-based, and location-based analysis**
- Includes both **operational and financial metrics**
- Enables creation of **KPIs and dashboards**
- Suitable for tools like Excel, MySQL, Python, and Power BI

Purpose of Dataset

The dataset is used to:

- Analyze passenger demand and travel patterns
- Evaluate financial performance (revenue & profit)
- Monitor operational efficiency (delays, occupancy)
- Build interactive dashboards for decision-making

Overall, the dataset provides a comprehensive foundation for performing end-to-end data analytics in the APSRTC bus transportation domain.

Data Collection

The data used in this project is a structured dataset representing the operations of the Indian Road Transport Corporation (RTC) bus system. The dataset was synthetically generated to simulate real-world transportation data and is designed to support data analytics learning and project development.

The dataset includes information related to bus trips, passengers, revenue, routes, stations, buses, and time. It follows a star schema structure, consisting of one fact table and multiple dimension tables, which enables efficient storage and analysis of large volumes of data.

Data collection in this project involves compiling and organizing data from multiple logical sources into a unified format. These sources include:

- **Trip-level operational data** such as passenger count, revenue, delays, and fuel costs
- **Station data** including station names, cities, and regional zones
- **Bus details** such as bus type, seating capacity, operator, and fuel type
- **Route information** including source, destination, distance, and duration
- **Date and time attributes** to support time-based analysis

The dataset was prepared in Microsoft Excel format and later imported into MySQL for data cleaning and transformation. Python was used to perform additional preprocessing and exploratory data analysis. Finally, the cleaned and structured data was loaded into Microsoft Power BI for visualization and dashboard creation.

Although the dataset is synthetic, it closely resembles real-world RTC operations and includes realistic variations in passenger demand, traffic conditions, weather impact, and seasonal trends. This ensures that the analysis performed in this project is meaningful and applicable to real-world scenarios.

Overall, the data collection process ensures that the dataset is comprehensive, structured, and suitable for performing end-to-end data analytics, from cleaning and transformation to visualization and insight generation.

Raw Data Description

The raw data used in this project represents the initial, unprocessed dataset of the Indian Road Transport Corporation (RTC) bus system. This data contains detailed information about bus operations, including trips, passengers, revenue, routes, stations, and time-related attributes. Before analysis, the raw data exists in its original form, which may include inconsistencies, missing values, and unstructured formats.

The dataset is organized into multiple tables following a star schema design, consisting of one central fact table and several dimension tables. Each table captures different aspects of the transportation system and is linked through unique identifiers.

Data Cleaning Process

Data cleaning is a critical step in the data analytics workflow, as the quality of analysis depends on the accuracy and consistency of the dataset. The raw RTC bus dataset contained issues such as missing values, duplicate records, inconsistent formats, and invalid entries. A systematic data cleaning process was performed using MySQL and Python to ensure the dataset was reliable and ready for analysis.

1. Handling Missing Values

The dataset contained NULL values in key columns such as passengers, revenue, and delay. These were handled using appropriate techniques:

- Missing numerical values were replaced using average (mean) values where applicable
- In cases where missing data was minimal, affected rows were removed
- Default values were used for certain fields (e.g., delay set to 0 where appropriate)

This ensured completeness without significantly distorting the dataset.

2. Removing Duplicate Records

Duplicate entries were identified based on key attributes such as trip_id, bus_id, and date_id. These duplicates were removed to prevent double-counting and inaccurate analysis.

- SQL queries were used to detect and delete duplicate rows
- Only unique records were retained in the final dataset

3. Handling Zero and Invalid Values

Certain records contained unrealistic or invalid values, such as:

- passengers = 0
- revenue = 0
- negative values in revenue or profit

These were addressed by:

- Removing rows with invalid values where necessary
- Replacing zero values with meaningful estimates (e.g., average values)
- Converting negative values to valid ranges where applicable

4. Standardizing Data Formats

Categorical data such as city names, bus types, and weather conditions contained inconsistencies (e.g., different cases or spellings).

- Text values were standardized using functions (e.g., UPPER, TRIM)
- Consistent naming conventions were applied across all tables

This ensured uniformity in grouping and analysis.

5. Date Formatting and Transformation

The date column was converted into a standard date format to enable time-based analysis.

- Extracted additional attributes such as day, month, year, and weekday
- Ensured consistency in date representation across the dataset

6. Data Type Corrections

Incorrect data types were identified and corrected:

- Numerical columns were converted to appropriate numeric formats
- Date columns were converted to datetime format
- Categorical fields were properly defined

7. Outlier Detection and Treatment

Extreme values (outliers) in columns such as delay_min, revenue, and passengers were analyzed:

- Identified using statistical summaries and visualization
- Treated by capping or removing unrealistic values

8. Feature Engineering

New columns were created to enhance analysis:

- Profit Margin = Profit / Revenue
- Revenue per Passenger
- Time-based features (month, year)

These derived metrics provided deeper analytical insights.

Data Transformation

Data transformation is a crucial step in preparing cleaned data for analysis and visualization. After completing the data cleaning process, the APSRTC bus dataset was transformed into a structured and analysis-ready format using MySQL and Python. The goal of this phase was to convert raw and inconsistent data into meaningful formats that support efficient querying, aggregation, and reporting.

One of the key transformations involved restructuring the dataset into a **star schema model**, consisting of a central fact table (fact_trips) and multiple dimension tables (dim_stations, dim_buses, dim_routes, and dim_dates). This transformation improved data organization and enabled multidimensional analysis across different business perspectives such as time, location, and operations.

Data type conversions were performed to ensure compatibility with analytical tools. Numerical fields such as revenue, profit, passengers, and fuel cost were converted into appropriate numeric formats, while date fields were transformed into standard datetime formats. This allowed accurate calculations and time-based analysis.

New derived columns were created to enhance analytical capabilities. These include:

- **Profit Margin** (profit ÷ revenue), which measures operational efficiency
- **Revenue per Passenger**, which helps evaluate pricing and demand
- **Time-based attributes** such as day, month, year, and weekday extracted from the date column

Categorical data was also transformed for consistency and usability. Columns such as bus type, operator, weather, and traffic level were standardized and encoded where necessary to ensure uniform analysis and grouping.

Aggregation and summarization transformations were applied to support reporting and dashboard creation. Data was grouped by key dimensions such as city, route, bus type, and time period to calculate metrics like total revenue, total profit, average occupancy, and passenger counts.

Additionally, indexing and key relationships were established between fact and dimension tables using unique identifiers. This ensured referential integrity and enabled efficient joins during analysis.

The transformed dataset was then exported into formats suitable for further analysis in Python and visualization in Microsoft Power BI. These transformations enabled seamless integration across tools and improved performance in dashboard development.

In summary, the data transformation process converted cleaned data into a structured, enriched, and analysis-ready format. This step played a vital role in enabling accurate insights, efficient reporting, and effective decision-making in the APSRTC bus data analytics project.

Star Schema Design

The star schema is a widely used data modeling technique in data warehousing that organizes data into a central fact table connected to multiple dimension tables. In this project, a star schema design was implemented to efficiently structure the RTC bus dataset for analysis and reporting.

At the core of the schema is the **fact table (fact_trips)**, which contains transactional data at the trip level. Each record in this table represents a single bus trip and includes measurable metrics such as passengers, revenue, fuel cost, delay time, occupancy rate, and profit. The fact table also includes foreign keys that link to the dimension tables, enabling detailed analysis across multiple perspectives.

Surrounding the fact table are four dimension tables:

- **dim_stations**: Contains information about bus stations, including station name, city, state, and zone. This table supports location-based analysis.
- **dim_buses**: Stores details about buses such as bus type, operator, seating capacity, and fuel type, enabling analysis of performance across different bus categories.
- **dim_routes**: Includes route-related information such as source, destination, distance, and duration, allowing route-level performance evaluation.
- **dim_dates**: Provides time-based attributes such as date, day, month, year, and weekday, supporting time-series analysis and trend identification.

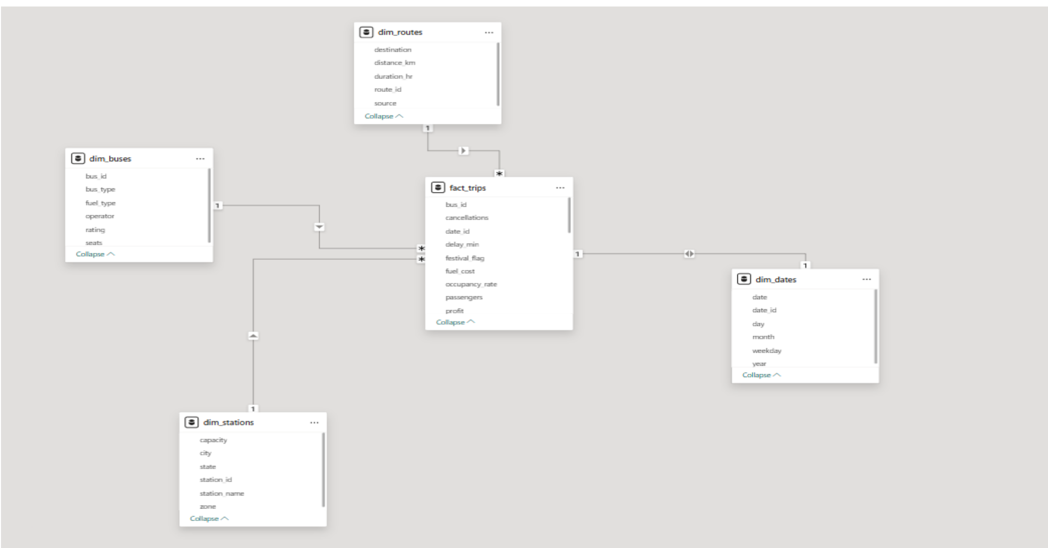
The relationships between the fact table and dimension tables are established using unique identifiers (primary and foreign keys). Each dimension table is linked to the fact table in a one-to-many relationship, forming a star-like structure.

This design offers several advantages:

- **Improved Query Performance**: Simplified joins enable faster data retrieval
- **Ease of Use**: Logical structure makes it easy for analysts to understand and query data
- **Scalability**: New dimensions or metrics can be added without affecting existing structure
- **Enhanced Analytics**: Supports multidimensional analysis across time, location, routes, and operations

The star schema design plays a critical role in this project by providing a strong foundation for data analysis in MySQL, Python, and visualization in Microsoft Power BI. It enables efficient aggregation, filtering, and slicing of data, which is essential for generating meaningful insights and building interactive dashboards.

In summary, the implementation of the star schema ensures that the APSRTC dataset is well-structured, optimized for performance, and suitable for advanced data analytics and business intelligence applications.



Fact Table Description

The fact table is the central component of the star schema design, containing quantitative and transactional data used for analysis. In this project, the **fact_trips** table serves as the core dataset, capturing detailed information about each bus trip in the RTC system. Every record in this table represents a single trip and includes measurable metrics that support performance evaluation and decision-making.

The fact_trips table is designed at the **granularity of one row per trip**, ensuring that all calculations and aggregations can be performed accurately at different levels such as daily, monthly, route-wise, or city-wise analysis. It contains both **foreign keys** linking to dimension tables and **measures** used for analytical computations.

Dimension Tables Description

Dimension tables are an essential component of the star schema design, providing descriptive and contextual information that supports detailed analysis of the fact table. In this project, four dimension tables—dim_stations, dim_buses, dim_routes, and dim_dates—are used to enrich the transactional data in the fact_trips table and enable multidimensional analysis.

1. dim_stations

The dim_stations table contains information related to bus stations and their geographical attributes. It enables location-based analysis of passenger demand and operational performance.

Key Columns:

- station_id – Unique identifier for each station
- station_name – Name of the station
- city – City where the station is located
- state – State name
- zone – Regional classification (North, South, East, West)
- capacity – Station capacity

Purpose:

- Analyze performance by city or region
- Identify high-demand locations
- Support geographic comparisons

2. dim_buses

The dim_buses table provides details about the buses used in operations. It supports analysis based on bus characteristics and performance.

Key Columns:

- bus_id – Unique identifier for each bus
- bus_type – Type (AC, Non-AC, Sleeper, etc.)
- operator – Operating authority (e.g., APSRTC, TSRTC)
- seats – Seating capacity
- fuel_type – Diesel or Electric
- rating – Bus quality rating

Purpose:

- Compare performance across bus types
- Evaluate profitability by operator
- Analyze impact of fuel type on costs

3. dim_routes

The dim_routes table contains route-specific information, enabling route-level performance evaluation.

Key Columns:

- route_id – Unique route identifier
- source – Starting location
- destination – Ending location
- distance_km – Distance of the route
- duration_hr – Travel duration

Purpose:

- Analyze passenger and revenue trends by route
- Identify high-performing and underperforming routes
- Evaluate efficiency based on distance and duration

4. dim_dates

The dim_dates table provides time-related attributes that support time-series analysis and trend identification.

Key Columns:

- date_id – Unique identifier for each date
- date – Actual date
- day – Day of the month
- month – Month
- year – Year
- weekday – Day of the week

Purpose:

- Perform time-based analysis (daily, monthly, yearly trends)
- Identify seasonal patterns and trends
- Analyze performance during weekdays vs weekends or festivals

Relationship Modeling

Relationship modeling defines how different tables in a database are connected to enable efficient data analysis and reporting. In this project, relationships are established between the central fact table (fact_trips) and the dimension tables (dim_stations, dim_buses, dim_routes, and dim_dates) using a star schema design.

The fact_trips table acts as the central hub, containing transactional data at the trip level. Each record in this table is linked to the corresponding dimension tables through foreign keys. These relationships allow the integration of descriptive attributes from dimension tables with measurable data from the fact table.

Types of Relationships

The relationships in this model are primarily **one-to-many (1:N)**:

- One record in a dimension table can relate to multiple records in the fact table
- Each record in the fact table references exactly one record in each dimension table

For example:

- A single **station** can be associated with many trips
- A single **bus** can operate multiple trips
- A single **route** can have multiple trips over time
- A single **date** can include multiple trips

Key Relationships

- fact_trips.date_id → dim_dates.date_id
- fact_trips.bus_id → dim_buses.bus_id
- fact_trips.route_id → dim_routes.route_id
- fact_trips.station_id → dim_stations.station_id

These relationships are established using **primary keys in dimension tables** and **foreign keys in the fact table**, ensuring referential integrity.

Relationship Direction and Filtering

In analytical tools such as Microsoft Power BI, relationships are typically defined with **single-direction filtering** from dimension tables to the fact table. This allows users to filter data based on attributes such as city, bus type, route, or date, and view the corresponding impact on metrics like revenue, passengers, and profit.

Benefits of Relationship Modeling

- Enables **data integration** across multiple tables
- Supports **multidimensional analysis** (time, location, routes, buses)
- Improves **query performance** and efficiency
- Ensures **data consistency and integrity**
- Simplifies dashboard creation and filtering

Excel and Its Usage

Microsoft Excel plays an important role in the initial stage of data analysis by providing a simple and effective environment for organizing and presenting data. In this project, Excel was used not only for data exploration but also for enhancing data readability through proper formatting techniques such as header styling, color formatting, and border design.

Header Formatting

The header row is the most important part of any dataset as it defines the meaning of each column. In this project:

- The header row was highlighted using **bold font** to distinguish it from the data
- A **larger font size** was applied to improve visibility

- Headers were aligned centrally for a clean and professional look
- Column names were made clear and consistent for better understanding

This formatting ensures that users can easily identify and interpret each column in the dataset.

Color Formatting

Color formatting was applied to improve readability and visual appeal:

- A **background color** (such as light blue or gray) was applied to the header row
- Alternate row shading (banded rows) was used to make large datasets easier to read
- Important columns such as revenue, profit, and passengers were highlighted using subtle colors
- Conditional formatting was applied to:
  - Highlight high values (e.g., high revenue or profit)
  - Identify low or negative values
  - Detect anomalies in data

Color formatting helps in quickly identifying patterns and trends without complex analysis.

Border Formatting

Borders were used to clearly separate data and improve structure:

- **Outer borders** were applied to define the dataset boundary
- **Inner gridlines** were added to separate rows and columns
- Header rows were given **thicker borders** to distinguish them from data rows
- Consistent border styles were used throughout the sheet for a clean appearance

This ensures that the dataset looks organized and easy to navigate.

Additional Formatting Enhancements

- Column widths were adjusted to fit content properly
- Text wrapping was applied where necessary
- Number formatting was used for currency (revenue, profit) and percentages (occupancy rate)
- Filters were enabled on headers to allow quick sorting and searching

Importance of Excel Formatting

Proper formatting in Excel:

- Improves data readability and usability
- Helps in identifying trends and outliers quickly
- Enhances presentation quality for reports
- Makes the dataset ready for further processing in MySQL, Python, and Power BI

MySQL and Its Usage

MySQL is a widely used relational database management system (RDBMS) that plays a crucial role in storing, managing, and processing structured data. In this project, MySQL is used as the primary platform for data storage, cleaning, transformation, and querying of the RTC bus dataset.

Role of MySQL in the Project

MySQL is utilized to manage the dataset designed in a star schema format, consisting of one fact table (fact_trips) and multiple dimension tables (dim_stations, dim_buses, dim_routes, and dim_dates). It serves as the backbone for handling large volumes of structured data efficiently.

Key Uses of MySQL

1. Data Import and Storage

- The dataset is initially stored in Excel format and then imported into MySQL tables
- Separate tables are created for fact and dimension data
- Proper data types and constraints are defined during table creation

2. Data Cleaning

MySQL is used to clean and prepare the dataset by:

- Removing duplicate records using SQL queries
- Handling NULL values with functions like IFNULL()
- Eliminating zero and invalid values
- Standardizing text data using functions like UPPER() and TRIM()

3. Data Transformation

- Creation of derived columns such as profit margin and revenue per passenger
- Extraction of date components (day, month, year)
- Aggregation of data using GROUP BY for analysis

4. Data Integration (Joins)

MySQL enables combining data from multiple tables using JOIN operations:

- INNER JOIN to merge fact and dimension tables
- Enables analysis across routes, cities, buses, and time

5. Data Analysis Using SQL Queries

Various analytical queries are executed, such as:

- Total revenue and profit by city or route
- Passenger trends over time
- Delay analysis based on traffic conditions
- Performance comparison across bus types

6. Performance Optimization

- Indexing is applied on key columns (IDs) to improve query speed
- Efficient schema design ensures faster data retrieval

Example Query

```
SELECT city, SUM(revenue) AS total_revenue
FROM fact_trips ft
JOIN dim_stations ds ON ft.station_id = ds.station_id
GROUP BY city;
```

This query calculates total revenue by city using a join between fact and dimension tables.

Benefits of Using MySQL

- Efficient handling of large datasets
- Strong support for structured querying (SQL)

- Easy integration with Python and Power BI
- Reliable data storage and management
- Enables complex analytical operations

Python and Its Usage

Python is a powerful and widely used programming language in the field of data analytics due to its simplicity, flexibility, and extensive library support. In this project, Python is used for data cleaning, transformation, exploratory data analysis (EDA), and visualization of the RTC bus dataset.

Role of Python in the Project

Python acts as a key analytical tool that bridges raw data and meaningful insights. It is primarily used after data is stored and partially cleaned in MySQL, enabling deeper analysis and visualization before building dashboards.

Key Libraries Used

- **pandas – For data manipulation and analysis**
- **numpy – For numerical operations**
- **matplotlib – For data visualization and chart creation**

These libraries provide efficient tools for handling structured data and performing complex analytical tasks.

Data Loading and Preparation

The dataset is loaded into Python using pandas, typically from CSV or Excel files. Data is then inspected using functions like head(), info(), and describe() to understand its structure and identify issues.

Data Cleaning in Python

Python is used to further clean and refine the dataset by:

- **Removing missing (NULL) values using dropna()**
- **Filtering out zero or invalid values**
- **Converting data types (e.g., string to datetime)**
- **Standardizing column formats**

This ensures the dataset is accurate and ready for analysis.

Feature Engineering

New columns are created to enhance analysis:

- Profit Margin = Profit ÷ Revenue
- Revenue per Passenger
- Time-based features such as year, month, and day

These derived features provide deeper insights into performance.

Exploratory Data Analysis (EDA)

Python is used to explore patterns and relationships in the data through:

- Aggregation using groupby()
- Statistical summaries
- Correlation analysis

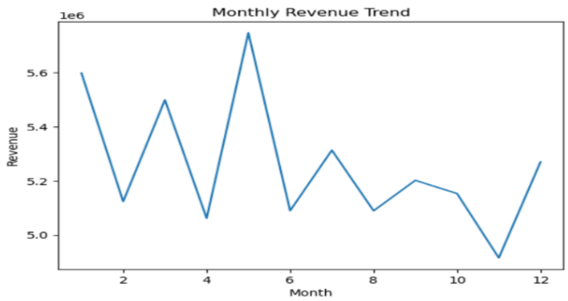
EDA helps in identifying trends, anomalies, and key factors affecting performance.

Data Visualization

Various charts are created using matplotlib, including:

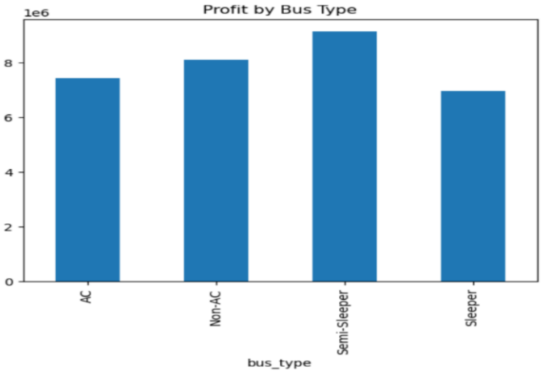
- Line charts (revenue trends over time)

```
[24]: df.groupby('month')['revenue'].sum().plot()
      plt.title("Monthly Revenue Trend")
      plt.xlabel("Month")
      plt.ylabel("Revenue")
      plt.show()
```



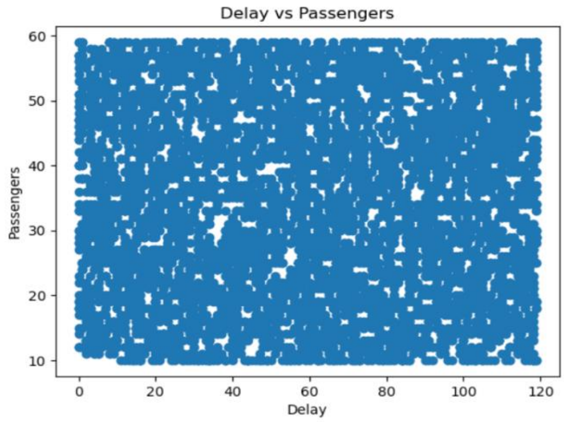
- **Bar charts (profit by bus type, revenue by city)**

```
[26]: df.groupby('bus_type')['profit'].sum().plot(kind='bar')
plt.title("Profit by Bus Type")
plt.show()
```



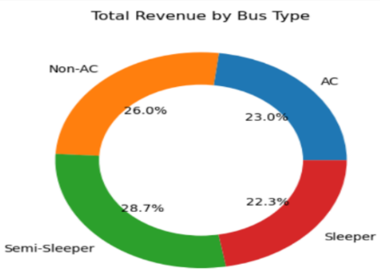
- **Scatter plots (delay vs passengers)**

```
[31]: plt.scatter(df['delay_min'], df['passengers'])
plt.xlabel("Delay")
plt.ylabel("Passengers")
plt.title("Delay vs Passengers")
plt.show()
```



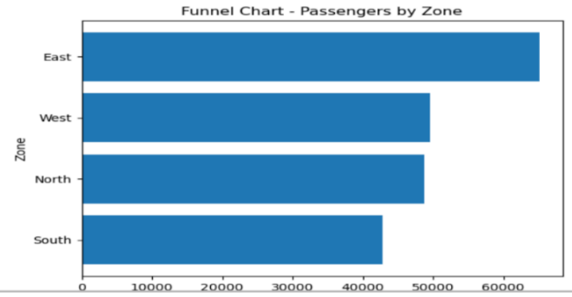
- **Donut charts (revenue distribution)**

```
[34]: plt.figure()

plt.pie(revenue_data, labels=revenue_data.index, autopct='%1.1f%%')

# Create donut shape
centre_circle = plt.Circle((0,0), 0.70, fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.title("Total Revenue by Bus Type")
plt.show()
```



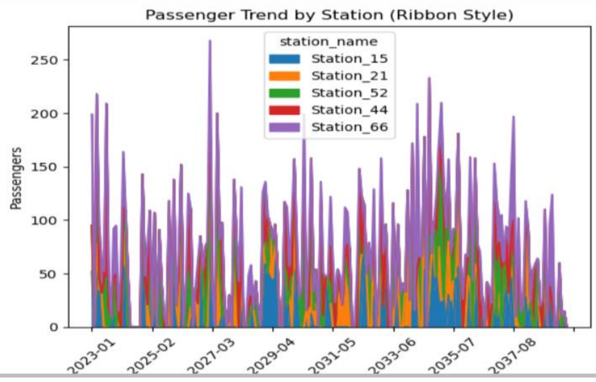
- **Funnel and ribbon-style charts for advanced insights**

```
plt.barh(zone_data.index, zone_data.values)

# Reverse order for funnel effect (largest on top)
plt.gca().invert_yaxis()

plt.title("Funnel Chart - Passengers by Zone")
plt.xlabel("Passengers")
plt.ylabel("Zone")

plt.show()
```

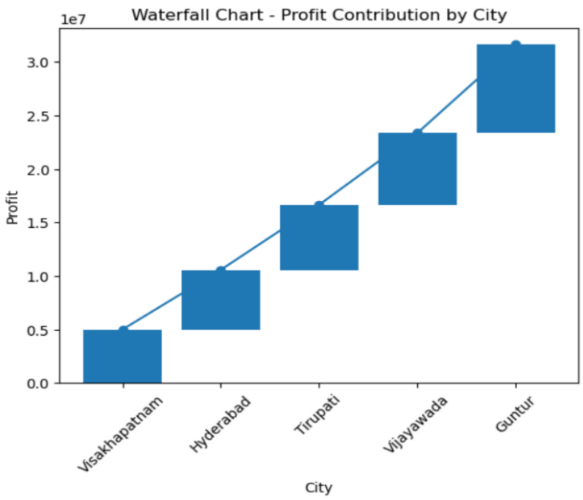


- **Ribbon**

```
[60]: pivot.plot.area()

plt.title("Passenger Trend by Station (Ribbon Style)")
plt.xlabel("Month")
plt.ylabel("Passengers")
plt.xticks(rotation=45)

plt.show()
```

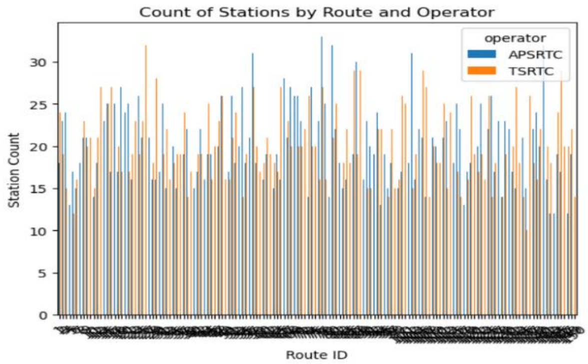


- **Water fall chart**

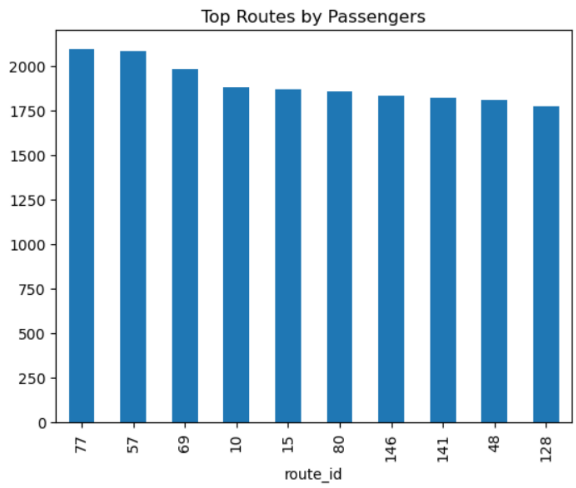


- **Bar and column chats**

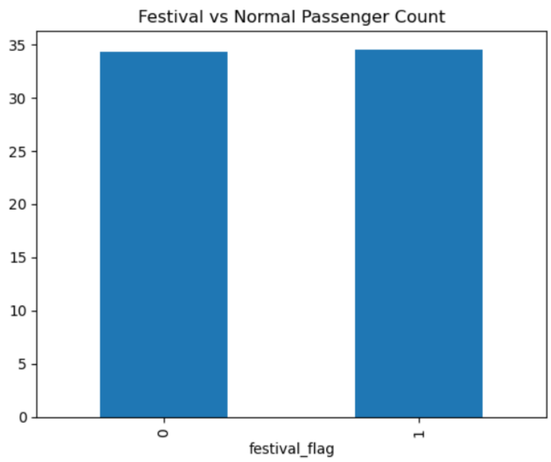
```
[52]: pivot.plot(kind='bar')

      plt.title("Count of Stations by Route and Operator")
      plt.xlabel("Route ID")
      plt.ylabel("Station Count")
      plt.xticks(rotation=45)

      plt.show()
```



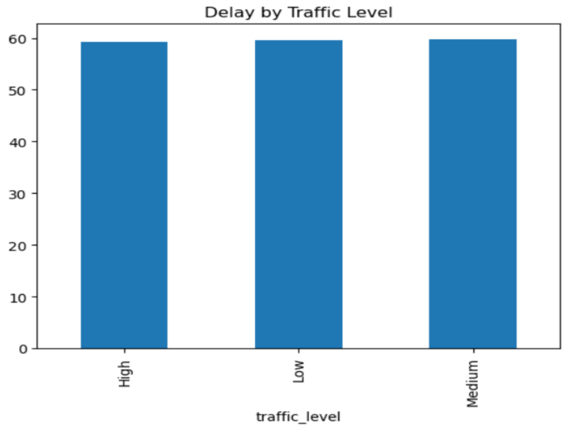

```
[27]: df.groupby('route_id')['passengers'].sum().sort_values(ascending=False).head(10).plot(kind='bar')
      plt.title("Top Routes by Passengers")
      plt.show()
```




```
[30]: df.groupby('festival_flag')['passengers'].mean().plot(kind='bar')
      plt.title("Festival vs Normal Passenger Count")
      plt.show()
```

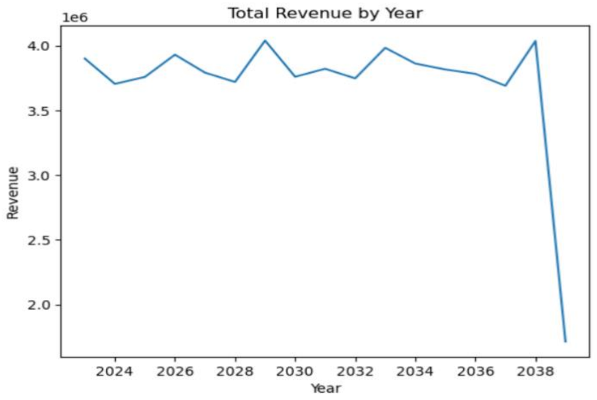


```
[28]: df.groupby('traffic_level')['delay_min'].mean().plot(kind='bar')
plt.title("Delay by Traffic Level")
plt.show()
```

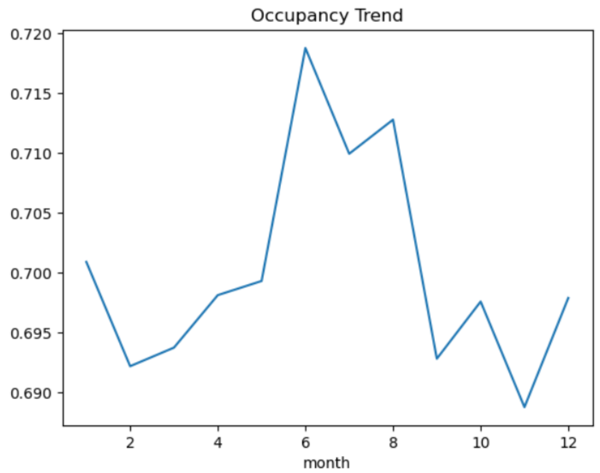


- **Line and Trands**

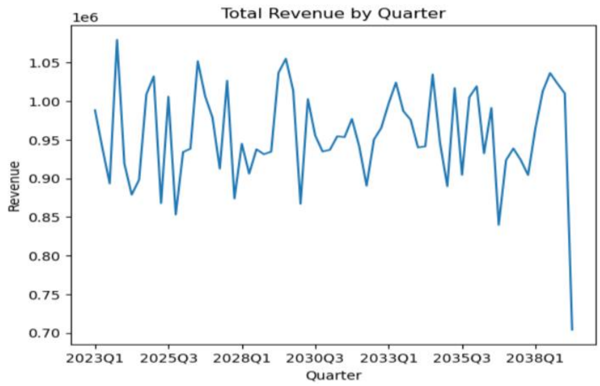
```
[43]: yearly = df.groupby('year')['revenue'].sum()

plt.figure()
yearly.plot()
plt.title("Total Revenue by Year")
plt.xlabel("Year")
plt.ylabel("Revenue")
plt.show()
```



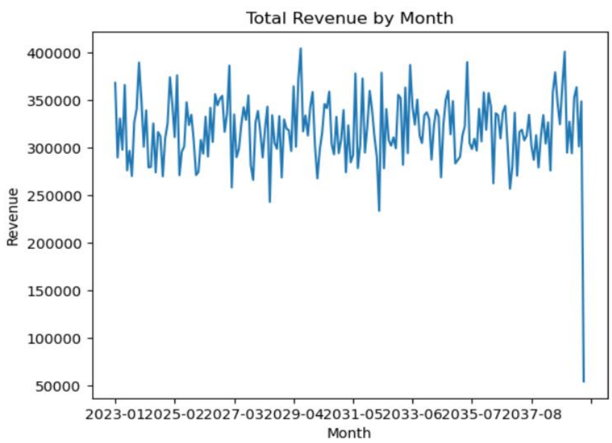
```
[29]: df.groupby('month')['occupancy_rate'].mean().plot()
plt.title("Occupancy Trend")
plt.show()
```




```
[44]: quarterly = df.groupby('quarter')['revenue'].sum()

plt.figure()
quarterly.plot()
plt.title("Total Revenue by Quarter")
plt.xlabel("Quarter")
plt.ylabel("Revenue")
plt.show()
```




```
[45]: monthly = df.groupby('month')['revenue'].sum()

plt.figure()
monthly.plot()
plt.title("Total Revenue by Month")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.show()
```



These visualizations help in understanding data intuitively.

Benefits of Using Python

- **Handles large datasets efficiently**
- **Supports advanced data analysis and visualization**
- **Easy integration with MySQL and Power BI**
- **Enables automation and reproducibility**
- **Provides flexibility for custom analysis**

Power BI and Its Usage

Microsoft Power BI is a powerful business intelligence and data visualization tool used to transform structured data into interactive dashboards and meaningful insights. In this project, Power BI is used as the final stage of the data analytics workflow to visualize and communicate findings derived from the RTC bus dataset.

Role of Power BI in the Project

Power BI serves as the primary platform for creating interactive dashboards that present key performance indicators (KPIs), trends, and patterns. It integrates data from MySQL and Python, enabling users to explore and analyze data dynamically.

Data Loading and Modeling

- **Cleaned and transformed data is imported into Power BI from MySQL or CSV files**
- **Relationships are established between fact and dimension tables based on the star schema design**
- **Data modeling ensures proper connections between tables for accurate analysis**

Data Transformation (Power Query)

Power BI's Power Query Editor is used for additional data transformation tasks:

- **Filtering unnecessary columns**
- **Handling remaining missing values**
- **Creating calculated columns**
- **Changing data types**

This step ensures the dataset is optimized for reporting.

DAX (Data Analysis Expressions)

DAX is used to create calculated measures and KPIs, such as:

- **Total Revenue**
- **Total Profit**
- **Profit Margin (%)**
- **Total Passengers**
- **Average Occupancy Rate**

These measures allow dynamic calculations based on filters and slicers.

Visuals

Data visualization is a key component of this project, as it transforms complex datasets into clear and meaningful insights. Various visual elements were used to represent patterns, trends, and relationships in the RTC bus dataset. These visuals were primarily created using Microsoft Power BI and Python to support effective analysis and decision-making.

Types of Visuals Used

1. Line Charts

Line charts were used to display trends over time, such as:

- **Total revenue by year, quarter, month, and day**
- **Profit trends over time**

These charts help identify growth patterns, seasonal variations, and fluctuations in performance.

2. Bar and Column Charts

Bar and column charts were used for comparison analysis:

- **Revenue by city**
- **Profit by bus type**
- **Passenger count by route**

They provide a clear comparison between different categories.

3. Donut Chart

Donut charts were used to show proportional distribution:

- **Revenue contribution by bus type**

This helps in understanding the share of each category in total performance.

4. Funnel Chart

Funnel charts were used to represent hierarchical or stage-based data:

- **Passenger distribution by zone**

This visualization highlights concentration and drop-off patterns across regions.

5. Ribbon Chart

Ribbon charts were used to show ranking changes over time:

- **Passenger demand by station across months**

This helps track how rankings of stations change over different time periods.

6. Scatter Plot

Scatter plots were used to analyze relationships between variables:

- **Delay vs passengers**
- **Revenue vs occupancy rate**

These charts help identify correlations and patterns between metrics.

7. Area Chart

Area charts were used to display cumulative trends:

- **Occupancy rate over time**

They provide a visual representation of volume and progression.

8. Treemap

Treemaps were used for hierarchical data representation:

- **Passenger distribution by city**

This allows quick identification of dominant categories.

9. Matrix/Table Visual

Matrix visuals were used for detailed analysis:

- **City vs bus type with occupancy rate**

They provide structured and tabular insights with conditional formatting.

10. Waterfall Chart

Waterfall charts were used to visualize the cumulative impact of different categories on a total value:

- **Profit contribution by city**

This chart shows how individual cities add to or reduce overall profit, making it easier to identify high-performing and low-performing regions. It is especially useful for understanding step-by-step changes in financial metrics.

Interactive Features

- **Slicers for filtering data by date, city, bus type, and zone**
- **Drill-down and drill-through capabilities**
- **Cross-filtering between visuals**
- **Tooltip-based detailed insights**

Importance of Visuals

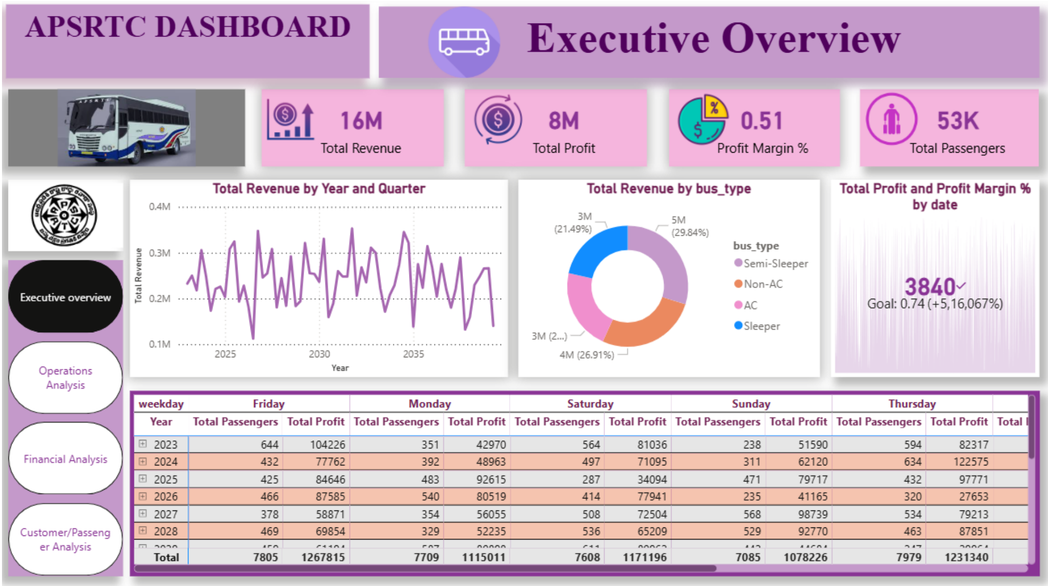
- **Simplifies complex data into understandable formats**
- **Helps identify trends, patterns, and anomalies**
- **Supports quick and effective decision-making**
- **Enhances user interaction and dashboard experience**

Dashboard Development

A multi-page interactive dashboard is created to present insights:

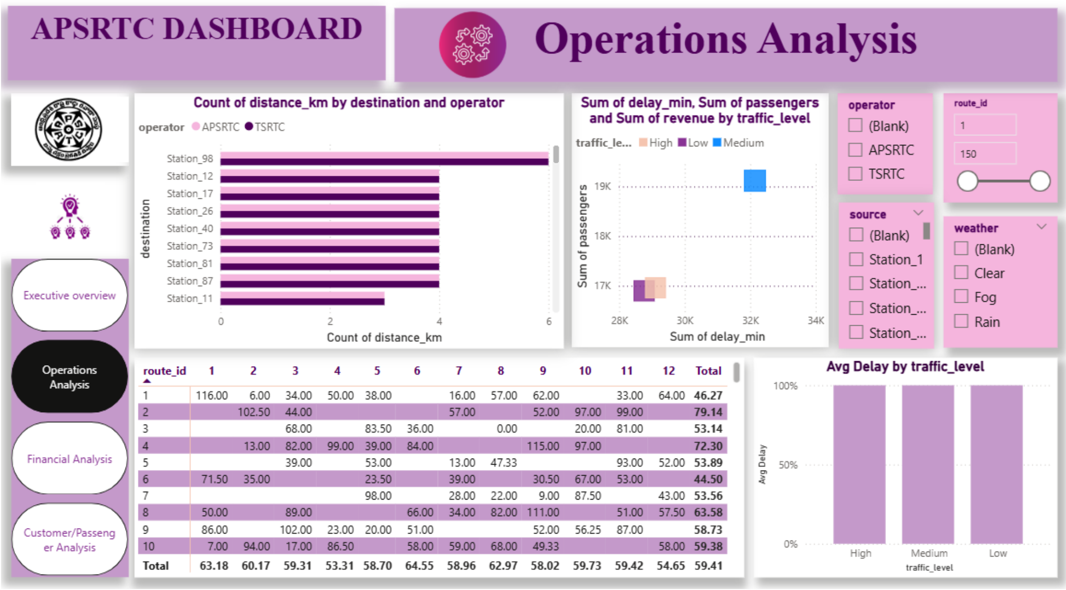
Page 1: Executive Overview

- **High-level KPIs and revenue trends**



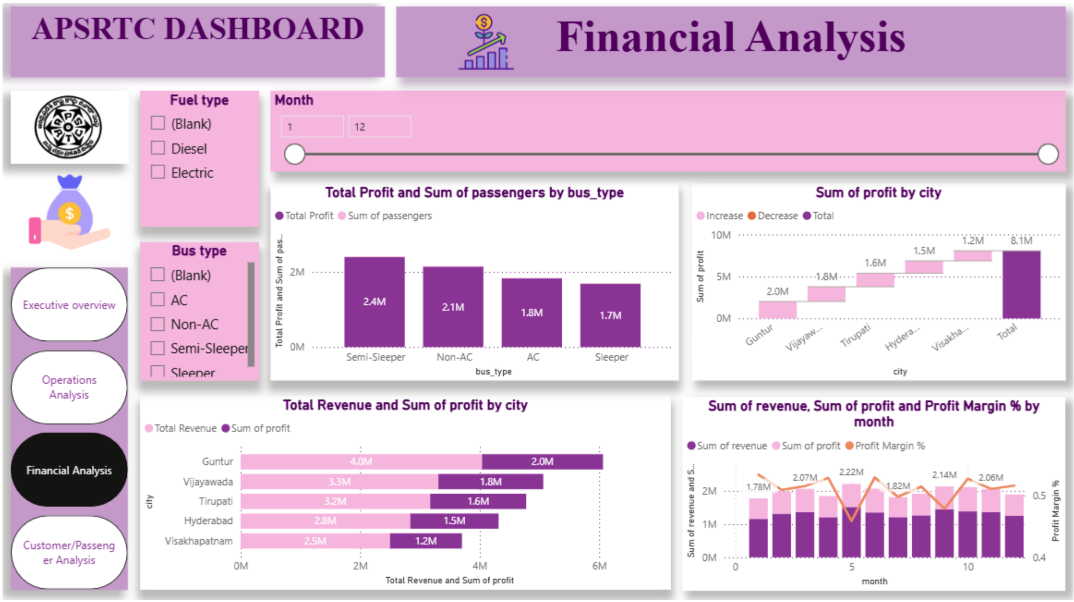
Page 2: Operations Dashboard

- **Delay analysis and route performance**



Page 3: Financial Dashboard

- **Revenue and profit analysis**



Page 4: Customer Dashboard

- **Passenger demand and behavior analysis**

Visualizations Used

- **Line charts (trends over time)**

- **Bar and column charts (comparisons)**
- **Donut charts (distribution)**
- **Funnel charts (zone-wise passenger flow)**
- **Ribbon charts (ranking changes over time)**
- **Scatter plots (relationship analysis)**
- **Waterfall chat**

Interactive Features

- **Slicers for filtering by date, city, bus type, and zone**
- **Drill-down and drill-through capabilities**
- **Tooltip pages for detailed insights**
- **Cross-filtering between visuals**

Benefits of Using Power BI

- **Interactive and user-friendly dashboards**
- **Real-time filtering and data exploration**
- **Integration with multiple data sources**
- **Advanced visualization capabilities**
- **Supports data-driven decision-making**

Challenges Faced

The development of this data analytics project involved multiple tools and stages, including Excel, MySQL, Python, and Microsoft Power BI. Throughout the process, several technical and analytical challenges were encountered. These challenges provided valuable learning opportunities and contributed to a deeper understanding of data handling, transformation, and visualization.

1. Challenges in Excel

Excel was used in the initial stage for data exploration and basic formatting. While it is user-friendly, several challenges were faced:

Data Size Limitation

Handling a dataset with multiple columns and thousands of rows made Excel slower and less efficient. Large datasets caused delays in loading, filtering, and applying formulas.

Data Inconsistency

The raw dataset contained:

- **Missing values**
- **Duplicate entries**
- **Inconsistent naming (e.g., “AC”, “ac”, “Ac”)**

Cleaning such inconsistencies manually in Excel was time-consuming and error-prone.

Formatting Issues

Ensuring consistent formatting across:

- **Headers**
- **Dates**
- **Numerical values (currency, percentage)**

required careful attention and repeated adjustments.

Limited Advanced Analysis

Excel has limitations in handling complex joins and large-scale aggregations, which made it difficult to perform advanced data analysis.

2. Challenges in MySQL

MySQL was used for data storage, cleaning, and transformation. While powerful, several challenges were encountered:

Data Import Issues

- **Errors during importing CSV/Excel files**
- **Incorrect data types assigned to columns**
- **Encoding issues in text fields**

Handling NULL and Invalid Values

Managing missing and zero values required careful query design using functions like IFNULL() and conditional filtering.

Writing Complex Queries

- **Writing JOIN queries across multiple tables was initially complex**
- **Understanding relationships between fact and dimension tables required practice**

Performance Issues

- **Queries on large datasets were slow without indexing**
- **Optimization techniques such as indexing and query restructuring were needed**

Data Integrity

Maintaining referential integrity between fact and dimension tables required proper key constraints and validation.

3. Challenges in Python

Python was used for data cleaning, analysis, and visualization. The following challenges were faced:

File Handling Errors

- **File not found errors due to incorrect file paths**
- **Issues with reading CSV/Excel files**

Data Cleaning Complexity

- **Handling missing values and outliers programmatically**
- **Ensuring correct data types (especially date formats)**

Merging Data (Joins)

Combining multiple tables into a single dataset using pandas required:

- **Correct join keys**
- **Handling mismatched or missing records**

Visualization Limitations

- **Creating advanced charts like waterfall and funnel charts required custom code**
- **Managing large datasets in plots sometimes reduced readability**

Debugging Errors

- **Syntax errors and logical errors in code**
- **Understanding error messages and fixing issues took time**

4. Challenges in Microsoft Power BI

Power BI was used for dashboard creation and visualization. Several challenges were encountered:

Data Modeling Issues

- **Establishing correct relationships between tables**
- **Avoiding circular dependencies**
- **Ensuring proper filter direction**

DAX Complexity

- **Writing DAX formulas for KPIs such as profit margin and growth rates**
- **Understanding context (row context vs filter context)**

Performance Optimization

- **Large datasets caused slow dashboard performance**
- **Required optimization techniques such as reducing unnecessary columns**

Visual Design Challenges

- **Selecting appropriate visuals for different types of data**
- **Avoiding clutter and maintaining a clean dashboard layout**
- **Ensuring consistency across multiple pages**

Slicers and Interactivity

- **Managing slicer synchronization across pages**
- **Ensuring proper cross-filtering behavior**

5. Integration Challenges Across Tools

One of the major challenges was integrating data across different tools:

- **Moving data from Excel → MySQL → Python → Power BI**
- **Ensuring consistency in data formats and structures**
- **Avoiding data loss or mismatch during transitions**

6. Learning and Adaptation Challenges

- **Understanding new tools and technologies simultaneously**
- **Managing the end-to-end workflow**
- **Debugging issues across different platforms**

Key Learnings from Challenges

- **Importance of data cleaning and preprocessing**
- **Efficient use of SQL for data management**
- **Power of Python for analysis and visualization**
- **Best practices in dashboard design using Power BI**
- **Handling real-world data issues and complexities**

AI Integration Scope

The integration of Artificial Intelligence (AI) into the APSRTC bus data analytics system presents significant opportunities to enhance decision-making, optimize operations, and improve customer experience. While the current project focuses on descriptive and exploratory analytics, there is substantial scope to extend it into advanced predictive and prescriptive analytics using AI techniques.

1. Demand Forecasting

AI models such as time series forecasting (ARIMA, LSTM) can be used to predict future passenger demand based on historical data. This enables:

- **Better route planning**
- **Efficient bus allocation**
- **Reduced overcrowding and underutilization**

2. Revenue and Profit Prediction

Machine learning algorithms (e.g., regression models) can be applied to predict revenue and profit based on factors such as:

- **Passenger trends**
- **Seasonal variations**
- **Traffic and weather conditions**

This helps in financial planning and pricing strategies.

3. Delay Prediction and Optimization

AI models can analyze historical delay data along with traffic and weather conditions to:

- **Predict potential delays**
- **Suggest optimal departure times**
- **Improve on-time performance**

4. Route Optimization

Using optimization algorithms and AI techniques:

- **Identify the most efficient routes**
- **Minimize travel time and fuel consumption**
- **Improve service coverage and efficiency**

5. Customer Segmentation

Clustering algorithms (e.g., K-Means) can be used to group passengers based on travel patterns:

- **Frequent travelers**
- **Occasional users**
- **Seasonal passengers**

This enables targeted services and personalized offers.

6. Anomaly Detection

AI can detect unusual patterns in data, such as:

- **Sudden drops in passenger count**
- **Abnormal revenue fluctuations**
- **Unexpected delays**

This helps in early issue detection and corrective actions.

7. Real-Time Analytics Integration

AI-powered systems can process real-time data from GPS, ticketing systems, and sensors to:

- **Provide live tracking of buses**
- **Predict arrival times**
- **Alert passengers about delays**

8. Chatbots and Customer Support

AI-based chatbots can be integrated to:

- **Provide real-time information to passengers**
- **Handle queries related to routes, schedules, and ticket availability**
- **Improve customer service efficiency**

9. Predictive Maintenance

AI can analyze vehicle performance data to:

- **Predict maintenance needs**
- **Reduce breakdowns**

- **Increase vehicle lifespan**

Benefits of AI Integration

- **Improved operational efficiency**
- **Enhanced decision-making through predictive insights**
- **Better customer satisfaction**
- **Cost optimization and increased profitability**
- **Real-time and automated analytics**

Project Evaluation

Project evaluation is an essential step to assess the effectiveness, accuracy, and overall success of the data analytics workflow. This evaluation focuses on how well the project achieved its objectives, the performance of the tools used, and the quality of insights generated.

1. Achievement of Objectives

The project successfully met its primary objectives:

- A structured **star schema data model** was designed and implemented
- Data cleaning and preprocessing were effectively performed using MySQL and Python
- Key performance indicators (KPIs) such as revenue, profit, passenger count, and occupancy rate were analyzed
- Interactive dashboards were developed using Microsoft Power BI

Overall, the project achieved its goal of transforming raw data into meaningful insights.

2. Data Quality and Reliability

After the data cleaning process:

- Missing values, duplicates, and inconsistencies were handled
- Data accuracy and consistency were significantly improved
- The dataset became reliable for analysis and reporting

However, since the dataset is synthetic, it may not fully capture real-world complexities.

3. Tool Performance Evaluation

Excel

- Effective for initial data exploration and formatting
- Limited in handling large datasets and advanced analysis

MySQL

- Efficient in managing structured data and performing complex queries
- Provided strong support for data cleaning and transformation

Python

- Highly effective for data analysis, feature engineering, and visualization
- Required debugging and coding expertise

Power BI

- Excellent for creating interactive dashboards and visualizations
- Enabled dynamic filtering and real-time insights

4. Analytical Effectiveness

The project provided valuable insights such as:

- Identification of high-performing routes and cities
- Understanding of passenger demand trends
- Analysis of factors affecting delays and profitability
- Evaluation of operational efficiency

These insights can support data-driven decision-making.

5. Dashboard Evaluation

The Power BI dashboard was evaluated based on:

- **Clarity**: Easy-to-understand visuals and layout
- **Interactivity**: Use of slicers, drill-down, and filters
- **Completeness**: Coverage of executive, operational, financial, and customer perspectives
- **Performance**: Optimized for smooth interaction

6. Limitations of the Project

- Use of synthetic dataset instead of real-world data

- Limited to descriptive and exploratory analytics
- No implementation of real-time data processing
- Advanced predictive modeling was not included

7. Overall Performance Assessment

The project demonstrates a successful end-to-end data analytics workflow:

- Data collection → Cleaning → Transformation → Analysis → Visualization

It effectively integrates multiple tools and techniques to deliver meaningful insights.

Conclusion

This project presents a comprehensive analysis of the Indian Andhra Pradesh Road Transport Corporation (APSRTC) bus system using a structured and multi-tool data analytics approach. The primary objective was to transform raw transportation data into meaningful insights that support operational efficiency, financial performance, and informed decision-making. By leveraging tools such as Excel, MySQL, Python, and Microsoft Power BI, the project successfully demonstrates an end-to-end data analytics workflow.

At the initial stage, data was explored and formatted using Excel, where proper structuring, header formatting, and basic validation ensured that the dataset was ready for further processing. The use of MySQL enabled efficient data storage, cleaning, and transformation. Issues such as missing values, duplicate records, and inconsistencies were resolved, ensuring high data quality and reliability. The implementation of a star schema model, consisting of a central fact table and multiple dimension tables, provided a strong foundation for structured analysis.

Python played a crucial role in performing exploratory data analysis (EDA) and advanced data manipulation. Through the use of libraries such as pandas, numpy, and matplotlib, the dataset was analyzed to identify key trends, patterns, and relationships. Feature engineering techniques, including the creation of metrics such as profit margin and revenue per passenger, enhanced the depth of analysis. Visualization in Python further supported the understanding of data through graphical representations.

The final stage of the project involved the development of an interactive dashboard using Microsoft Power BI. The dashboard was designed across multiple pages, including executive, operational, financial, and customer views. Various visualizations such as line charts, bar charts, donut charts, funnel charts, ribbon charts, and waterfall charts were used to present insights effectively. The use of DAX measures enabled dynamic calculations of KPIs such as total revenue, total profit, passenger count, and occupancy rate. Interactive features like slicers, drill-downs, and cross-filtering enhanced user experience and enabled detailed exploration of data.

The analysis revealed several important insights. Passenger demand was found to vary significantly across routes, cities, and time periods, with noticeable increases during festival seasons. Traffic conditions and weather were identified as key factors influencing delays and operational efficiency.

Certain bus types, particularly premium categories such as AC and sleeper buses, contributed more significantly to profitability. Additionally, fuel costs were observed to have a direct impact on overall profit margins. These insights highlight the importance of data-driven decision-making in optimizing transportation operations.

Despite its success, the project has certain limitations. The dataset used is synthetic and may not fully capture the complexities of real-world RTC operations. The analysis is primarily descriptive and exploratory, without the inclusion of advanced predictive or prescriptive models. Real-time data integration and automation were also beyond the scope of this project. However, these limitations provide opportunities for future enhancements.

The project also highlighted several challenges across different tools, including data inconsistencies in Excel, query optimization in MySQL, debugging in Python, and DAX complexity in Power BI. Overcoming these challenges contributed significantly to skill development and provided practical experience in handling real-world data scenarios.

Looking ahead, the project can be extended by integrating artificial intelligence and machine learning techniques. Predictive models can be developed for demand forecasting, delay prediction, and revenue estimation. Real-time analytics can be implemented using live data sources, and optimization algorithms can be applied to improve route planning and resource allocation. Additionally, customer-centric features such as personalized recommendations and AI-based chatbots can further enhance the system.

In conclusion, this project successfully demonstrates the application of data analytics in the transportation domain. It showcases how raw data can be transformed into actionable insights through systematic processes and modern analytical tools. The findings of this project can support stakeholders in improving operational efficiency, maximizing revenue, and enhancing passenger satisfaction. Overall, the project serves as a strong foundation for advanced analytics and intelligent transport solutions in the future.

Thank

You